

Datenbanken und Web-Techniken

Project Task

Open Data Integration with Accidents in Germany

Abteilung Datenmanagement
Prof. Dr. Michael Martin
Florian Hahn

Evaluation Subject DBW

Evaluation der Vorlesung 17:15 Uhr

<https://evaluation.tu-chemnitz.de/evasys/online.php?p=2V9JF>



Final Project - DBW

Schedule

- Introduction
- Motivation and Task
- Task Description
- Data Sources and Integration
- Examination
- Dates

Project focus

Official German open data
Multiple formats and interfaces
Own canonical schema
Own API with documentation
Examiner-verifiable answers

Final Project - DBW

Introduction

Open government data can be consumed through file downloads, map services and APIs.

Unfallatlas

- Accidents with personal injury
- CSV or Shapefile download
- WMS map service

Regionalatlas

- GeoJSON or Shapefile
- CSV basis data
- Embeddable atlas views

GENESIS

- RESTful/JSON API
- XLSX, CSV, XML

Govdata

- RESTful/JSON API
- XLSX, CSV, XML

This project focuses on integrating official German road accident data and regional statistics into a reusable own data service.

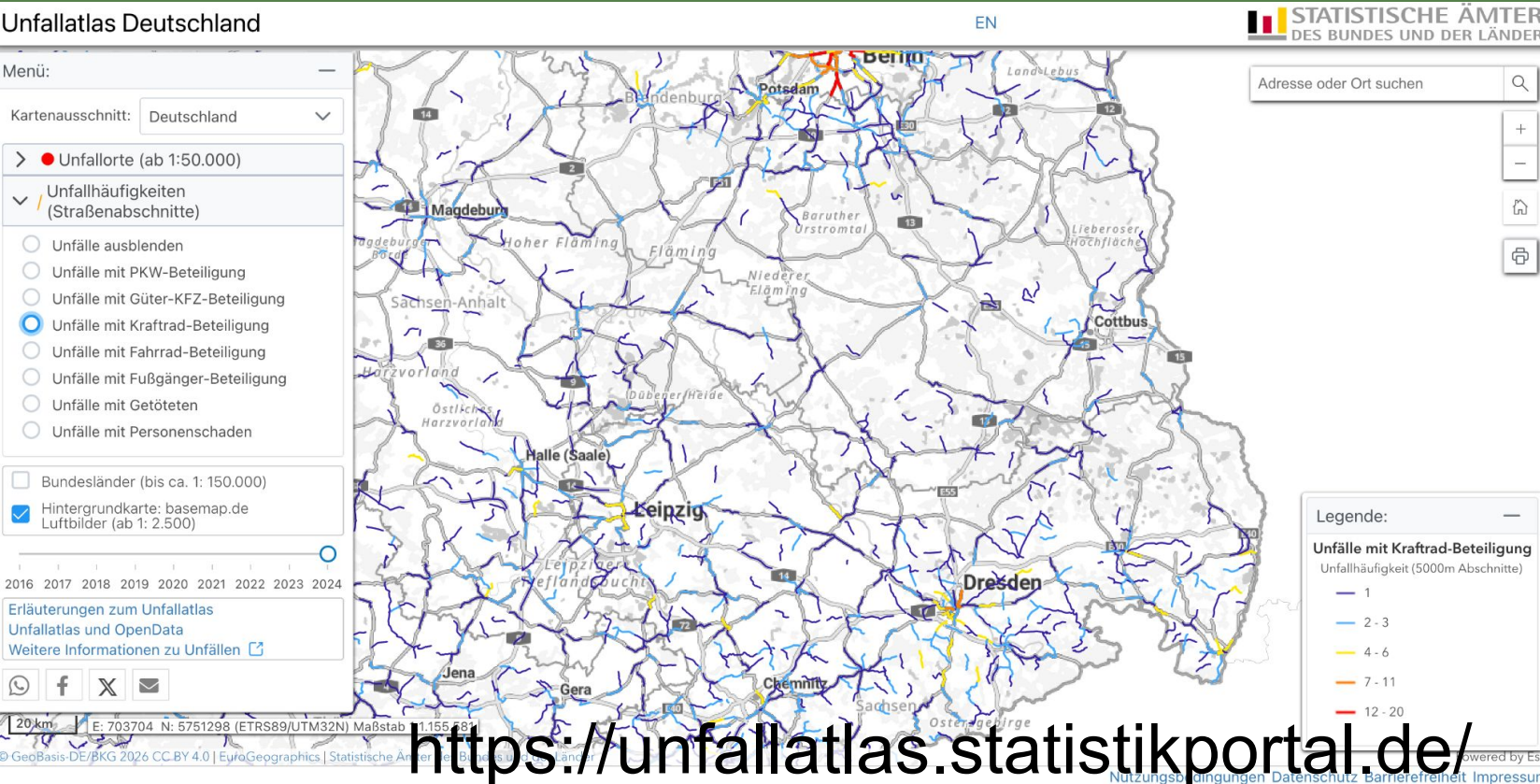
The goal is not only to download data, but to harmonise differing schemas, region keys and granularities and to provide the result through a documented student-built API.

It should be possible to answer different scenarios on your API endpoint and get reasonable answers.

Task Description - DBW

Einleitung

Unfallatlas Deutschland



Unfallatlas

oder
Format

daten

en
hten

GENESIS

- RESTful/JS
ON API
- XLSX, CSV,
XML

Govdata

- RESTful/JS
ON API
- XLSX, CSV,
XML

en und regionale Statistiken in einen

über eine dokumentierte, selbst entwickelte API bereitzustellen.

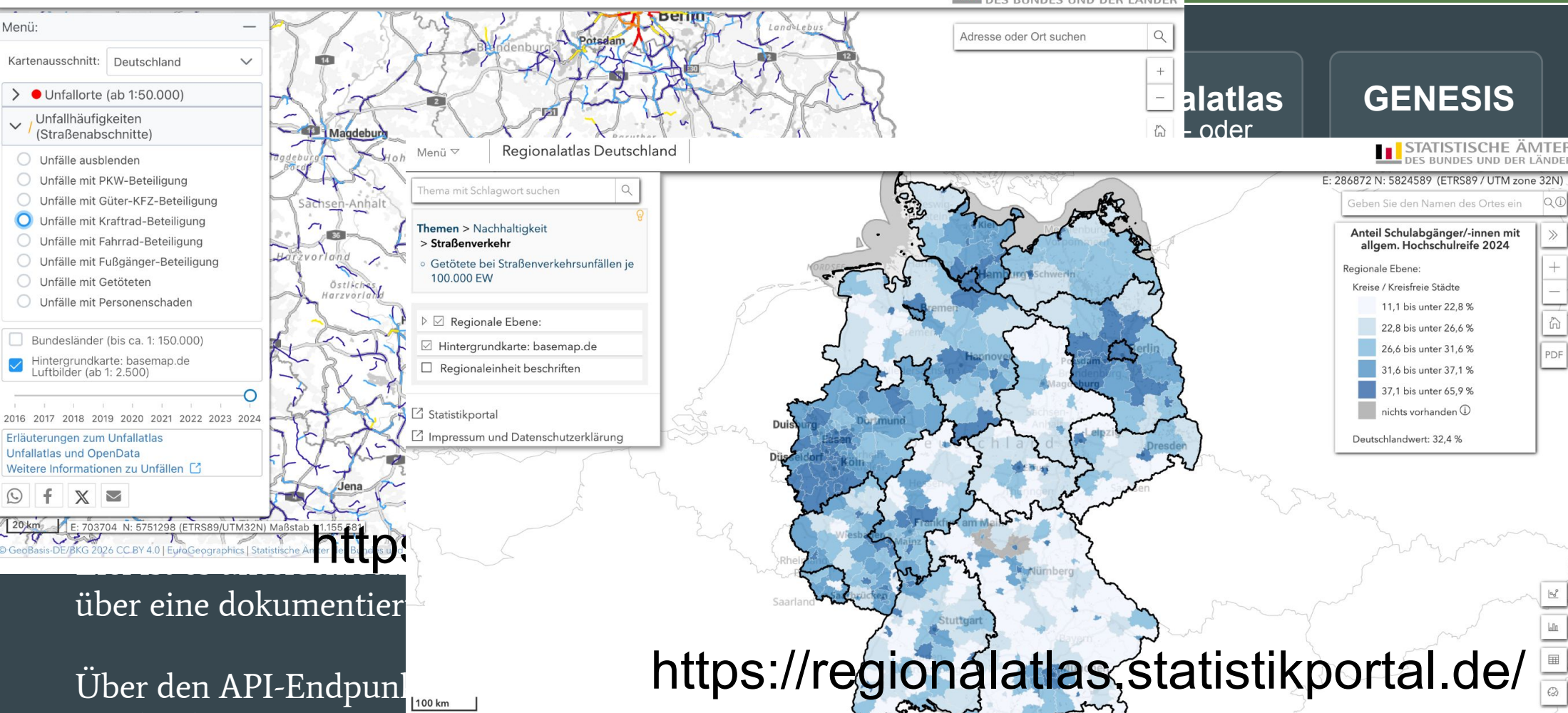
Über den API-Endpoint sollen verschiedene Szenarien abgefragt und plausible Antworten zurückgegeben werden können.

zu harmonisieren und das Ergebnis

Projektaufgabe - DBW

Einleitung

Unfallatlas Deutschland



über eine dokumentier

Über den API-Endpunkt
werden können.

<https://regionalatlas.statistikportal.de/>

Projektaufgabe - DBW

Einleitung

Unfallatlas Deutschland

Menü:

Kartenausschnitt: Deutschland

> ● Unfallorte (ab 1:50.000)

▼ Unfallhäufigkeiten (Straßenabschnitte)

☐ Unfälle ausblenden

☐ Unfälle mit PKW-Beteiligung

☐ Unfälle mit Güter-KFZ-Beteiligung

☒ Unfälle mit Kraftrad-Beteiligung

☐ Unfälle mit Fahrrad-Beteiligung

☐ Unfälle mit Fußgänger-Beteiligung

☐ Unfälle mit Getöteten

☐ Unfälle mit Personenschaden

☐ Bundesländer (bis ca. 1: 150.000)

☒ Hintergrundkarte: basemap.de Luftbilder (ab 1: 2.500)

2016 2017 2018 2019 2020 2021 2022 2023 2024

Erläuterungen zum Unfallatlas
Unfallatlas und OpenData
Weitere Informationen zu Unfällen

📄 📱 📧

Thema mit Schlagwort suchen

Themen > Nachhaltigkeit > Straßenverkehr

◦ Getötete bei Straßenverkehrsunfällen je 100.000 EW

☒ Regionale Ebene:

☒ Hintergrundkarte: basemap.de

☐ Regionaleinheit beschriften

☒ Statistikportal

☒ Impressum und Datenschutzerklärung

20 km

E: 703704 N: 5751298 (ETRS89/UTM32N) Maßstab: 1:155.581

© GeoBasis-DE/BKG 2026 CC-BY 4.0 | EuroGeographics | Statistische Ämter

über eine dokumentier

Über den API-Endpunkt
werden können.

Information und Technik
Nordrhein-Westfalen

OpenGeodata.NRW

unfallatlas

{...} JSON <> XML

Verkehrsunfälle mit Personenschaden in Deutschland (Unfallatlas)

• Lizenzinformation: Diese Daten stehen unter der [Datenlizenz Deutschland Namensnennung 2.0](#).

• Datensatzbeschreibung:

DSB_Unfallatlas_EN.pdf

DSB_Unfallatlas.pdf

• Metadaten im [Geoportal.NRW](#)

Punktgenaue Daten zu Straßenverkehrsunfällen in Deutschland für 2024

Datei	Letzte Änderung	Dateigröße
Unfallorte2024_EPSG25832_CSV.zip	09.07.2025 07:37:08	12.1 MB
Unfallorte2024_EPSG25832_Shape.zip	09.07.2025 07:37:08	26.9 MB

Punktgenaue Daten zu Straßenverkehrsunfällen in Deutschland für 2023

Datei	Letzte Änderung	Dateigröße
Unfallorte2023_EPSG25832_CSV.zip	17.05.2024 17:06:43	12.1 MB
Unfallorte2023_EPSG25832_Shape.zip	17.05.2024 17:06:43	27.0 MB

Punktgenaue Daten zu Straßenverkehrsunfällen in Deutschland für 2022

Datei	Letzte Änderung	Dateigröße
-------	-----------------	------------

https://www.opengeodata.nrw.de/produkte/transport_verkehr/unfallatlas/

<https://regionalatlas.statistikportal.de/>

ergebnis

en

Projektaufgabe - DBW

Ein

Unfallatlas

Menü:

Kartenausschl

> ● Unfall

Unfallh
(Straß

Unfälle

Unfälle

Unfälle

Unfälle

Unfälle

Unfälle

Unfälle

Unfälle

☐ Bundesli

☒ Hintergr
Luftbild

2016

2017

20

Erläuterungen

Unfallatlas ur

Weitere Infor

WhatsApp

f

20 km

GeoBasis-DE/B

Tabelle

Downloads:

XLSX

CSV

FLAT

XML

Optionen:

Q

→

↑

DIAGRAMM
ANZEIGEN

Infallatlas)

ung 2.0.

Letzte Änderung

Dateigröße

09.07.2025 07:37:08

12.1 MB

09.07.2025 07:37:08

26.9 MB

Letzte Änderung

Dateigröße

17.05.2024 17:06:43

27.0 MB

Letzte Änderung

Dateigröße

Jahr		Unfälle (insgesamt)	Unfälle mit Personenschaden	Schwerwiegende Unfälle mit Sachschaden i. e. S.	Son
Kreise und kreisfreie Städte		Anzahl	Anzahl	Anzahl	
2017					
DG	Deutschland	392 716	302 656	74 461	
01	Schleswig-Holstein	15 127	12 528	2 214	
01001	Flensburg, kreisfreie Stadt	423	368	43	
01002	Kiel, kreisfreie Stadt	1 290	1 141	111	
01003	Lübeck, kreisfreie Stadt, Hansestadt	1 307	1 132	141	
01004	Neumünster, kreisfreie Stadt	474	394	60	
01051	Dithmarschen, Kreis	649	539	99	
01053	Herzogtum Lauenburg, Kreis	858	695	142	
01054	Nordfriesland, Kreis	1 068	915	136	
01055	Ostholstein, Kreis	1 155	943	179	
01056	Pinneberg, Kreis				
01057	Plön, Kreis				
01058	Rendsburg-Eckernförde, Kreis				

<https://www.regionalstatistik.de/genesis/online?operation=abruftabelle&levelindex=1&levelid=1779366952256&auswahloperation=abruftabelleAuspraegungAuswaehlen&auswahlverzeichnis=ordnungssstruktur&auswahlziel=werteabruf&code=46241-01-03-4&auswahltext=&werteabruf=Werteabruf#abreadcrumb>

über eine dokumentier

Über den API-Endpunkt
werden können.

Final Project - DBW

Motivation and Task

In the context of regional development and optimisation, Open Data–based analyses are to be carried out. For this purpose, aggregates may be required for which an API or interface is needed.

The system to be developed by you must be able to answer these analyses reproducibly through an API.

Examples of examiner questions

- How many accidents occurred in Saxony in 2023?
- What is the earliest accident year in the data?
- Which districts had the highest accident rate per 100,000 inhabitants?

Core project outcome

- Integrated official open data
- Own integration script
- Own database schema
- Own API plus API documentation

Important principle

- No toy dataset
- No manual one-off copy and paste
- Answers must be derived from stored real data

Main task: To develop an API for at least three integrated datasets from the sources listed (page 15), such as the German Accident Atlas or the Regional Atlas, organised into defined categories (page 19).

Task Description

Preliminary Remarks

- All students in this semester have one practical project task.
- Detecting significant solution parts shared between submissions from different students is considered attempted fraud and will be marked as such for all students involved.
- The subtasks can be solved using any database management system, programming language, framework, library or deployment architecture.
- Supported languages are German and English.
- Main focus: correct data integration, schema design, API quality, documentation and reproducibility.

Task Description

Submission

- The submission consists of ONE ZIP archive named with the matriculation number.
- Include a PDF file of the term paper in A4 format.
- Include a separate ZIP archive with a maximum file size of 10 MiB containing: source code, database initialisation script, ETL/import scripts, API documentation export, and a short usage manual.
- The submission of the database itself is not required. The submission of an executable program is not required.
- Only one submission per student is required. The submission has to be on time.
- Please reduce large unused files and do not upload downloaded raw datasets if they can be reproduced from scripts.

Task Description

Term Paper conditions

- Write a term paper with about 5 pages of content, excluding cover, index, appendix, bibliography and API documentation.
- Keep a balanced ratio of text and figures. Screenshots only count if they are explained in the surrounding text.
- The cover page records the name, study course and matriculation number of the author.
- Give an overview of all utilized technologies and motivate why they were chosen for the corresponding subtask.
- Describe the complete architecture so that a reader understands all major parts without explicitly executing the project.

The term paper will be part of your first mark.

The term paper must explain

- source selection and access paths
- schema mapping and data harmonisation
- database design and indexing
- API design decisions
- update workflow and reproducibility
- limitations, plausibility checks and data quality issues

Task Description

Term Paper conditions

- All used sources, libraries and technologies are referenced. There are no restrictions on citation style, but styles should not be mixed.
- The appendix includes a complete API documentation. Listing examples only is not sufficient.
- For every endpoint, document methods, parameters, return values, data types, meaning and relevant HTTP status codes.
- For complex structures, also document the inner structure or schema of the returned object.
- Provide example requests and responses. OpenAPI or Swagger-based documentation is recommended.
- Document provenance and update behaviour: which external source was used, when it was retrieved and how it was mapped.

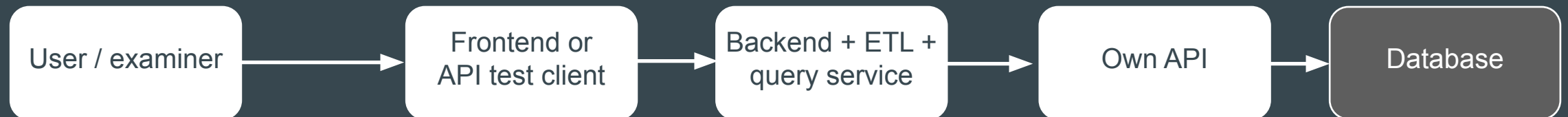
Task Description

Programming

External sources example

- Unfallatlas CSV or Shapefile
- Unfallatlas WMS
- Regionalatlas data files
- Regionaldatenbank or GENESIS API
- AGS or Gemeindeverzeichnis reference
- Govdata

The programme consists of an own database, a backend that processes and harmonises the data, and a web service API that exposes the stored result.
A lightweight frontend or API client for the demo is recommended.



Write an update script that refreshes the database from the official sources without abusing the external services.

Task Description

Data Sources and Interfaces

Unfallatlas

- Accidents with personal injury only
- Point data by year
- WMS map service
- Attributes include year, month, hour, weekday, category, light, participant flags and coordinates

Unfallatlas Data:

https://www.opengeodata.nrw.de/produkte/transport_verkehr/unfallatlas/

Unfallatlas Map:

<https://unfallatlas.statistikportal.de/>

Unfallatlas Number

explanation:

https://www.opengeodata.nrw.de/produkte/transport_verkehr/unfallatlas/DSB_Unfallatlas.pdf

Regionalatlas /Regionalstatistik Deutschland

- Geo data as GeoJSON or Shapefile
- Basis data as CSV
- Embeddable atlas views
- Useful for region geometries and thematic context

Regionalatlas:

<https://regionalatlas.statistikportal.de/>

Regionalstatistik:

https://www.regionalstatistik.de/genesis/online?operation=find&suchanweisung_language=de&query=unfall#abreadcrumb

Regionaldatenbank / GENESIS

- RESTful/JSON API or webservice
- Further formats: XLSX, CSV, XML
- Provides indicator values for regional comparison
- Use for population or further contextual statistics

Genesis:

<https://www-genesis.destatis.de/datenbank/online>

Example Saxony:

<https://www.statistik.sachsen.de/genonline/online>

AGS/ADM - GV-ISys

- Official region keys
- AGS is 8 digits
- Useful for harmonised joins between accident, geometry and indicator data

GV:ISys:

<https://www.destatis.de/DE/Themen/Laender-Regionen/Regionales/Gemeindeverzeichnis/inhalt.html#101366>

GovData

- Official Open Data Portal of Germany
- Scraping different sources with metdata and formats

“unfall key”:

<https://www.govdata.de/suche?tags=stra%C3%9Fenverkehrsunf%C3%A4lle>

Verkehrsunfall

<https://www.govdata.de/suche/daten/verkehrsunfalleb862c>

Minimum requirement: integrate at least three external data accesses (API or file) in total

Task Description

Data Integration and Harmonisation

- Different sources expose different schemas, region identifiers, temporal coverage and formats.
- Use official regional keys where possible and document your join strategy.
- The backend may keep raw import traces temporarily, but the final application data must live in the chosen database.
- Write an update script for new source deliveries and explain how duplicates or changed records are handled.



Mandatory quality aspects

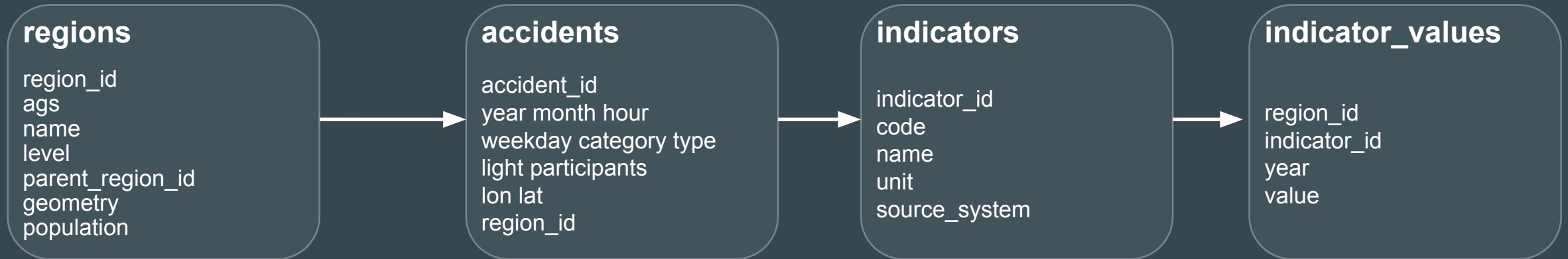
- reproducible imports
- documented source provenance
- plausibility checks
- clear explanation of limitations

Do not do this

- hand-editing answers in the database
- querying the live source on every frontend request
- ignoring source usage rules or creating avoidable load

Task Description

Database and Schema Suggestion



Schema requirements

- Students may choose any DBMS, but you must use either your own schema or the one suggested here.
- It is not sufficient to mirror source files 1:1 without modelling decisions.
- Separate events, regions, indicators and provenance in a clean way.

Task Description

Web Service Application Programming Interface

- The web service API provides access to the stored data through your backend. The frontend or test client is not allowed to communicate directly with the database.
- Provide endpoints for regions, filtered accidents, aggregated accident counts, metadata and provenance.
- API access failures are identified and responded to with meaningful error messages and suitable HTTP status codes.
- A full API documentation is mandatory. OpenAPI or Swagger is recommended.

```
{ "level": "state", "region": "Sachsen", "year":  
2023,  
  "accident_count": ...,  
  "source_snapshot": "..."} }
```

Example REST endpoints (GraphQL, SPARQL, SOAP, RPC are possible as well)

- GET /regions?level=state
- GET /accidents?state=SN&year=2023
- GET /aggregates/accidents?level=state&year=2023
- GET /aggregates/accidents?level=district&year=2024&category=1
- GET /metadata/sources
- GET /import-runs

Task Description

Query Categories

Counts and filters

- accidents by state, district or municipality
- filters by year, month, weekday or hour
- filters by accident category or participant flags

Time questions

- earliest available accident year overall
- earliest year per federal state
- year-over-year changes or trend summaries

Rates and rankings

- accident rate per 100,000 inhabitants
- top or bottom districts
- comparison across regions and years

Bonus: Zero cases

- municipalities with zero accidents in a year
- requires full region reference, not only event rows

Your own API must be able to answer questions from these categories using the stored data. The Zero cases category is bonus.

Task Description

Test Questions

- Mandatory questions (total: 5 + 2):
 - What is the earliest accident year in the complete dataset?
 - How many accidents involving personal injury occurred in Saxony in 2023?
 - From which year onwards is data available for North Rhine-Westphalia?
 - From which year onwards is data available for Mecklenburg-Western Pomerania?
 - How many accidents involving pedestrians occurred in Berlin in 2023?
- Questions that cannot be answered using only a single source dataset (at least 2 required), for example:
 - Traffic accidents per 100,000 registered passenger cars in a defined region
 - Other aggregations are also permitted.
- You are encouraged to formulate additional questions that can be answered by your service.
- For example:
 - Which five districts recorded the highest number of fatal accidents in 2024?
 - How many bicycle accidents occurred in Dresden in 2024?
 - Which municipalities in Saxony recorded zero reported accidents in 2023?
- How you implement the answering of these questions is up to you; however, a justification/explanation is mandatory.

Task Description

Test Questions

- Identify the respective licences under which the selected datasets may be processed and, if applicable, republished.
- Please include the corresponding provenance of the datasets in the database as well (either directly with the dataset or centrally) and transmit the relevant licences to the clients together with the responses.

Task Description

Frontend / Demo Client

The main assessment target is the data platform and API.

A small query client is strongly recommended for the live demo, e.g. browser, desktop app, mobile app, web app, terminal.

Recommended features: preset example queries, map or chart view, source metadata, spatial queries, caching (choose a feasible caching strategy) and an endpoint explorer.

Task Description

Evaluation

Grading rubric

- Total: 100 points:
- 25% Data integration and reproducibility
- 20% Schema and database modelling
- 20% API functionality and correctness
- 15% API documentation
- 10% Data quality and plausibility checks
- 10% Live demonstration and answerability of examiner queries
- Optional Bonus points from bonus category or bonus questions

Minimum expectations

- official German open data is used
- own canonical schema is implemented
- data are stored in a database of your choice
- your own API is documented
- examiner queries can be answered from real stored data

Task Description

Examination

- The examination consists of a 10-minute presentation. The focus is on the live demonstration rather than on reading slides.
- Briefly introduce the project, its development process, important design decisions and the used technologies.
- The main part of the presentation is to demonstrate your features and functionalities, especially the integrated data model and your API.
- Afterwards, a few questions are asked, focusing primarily on the project and the term paper, but they may also cover lecture content.
- Be prepared to answer a few examiner queries live by using your own interface or API client.

Task Description

Dates

- Handout of task description: 21.05.2026
- Submission deadline: 25.06.2026 23:59 via OPAL
- Oral exam and presentation slots: 02.07 – 16.07.2026
- Appointment allocation and room information: 1/336.2
- Enrollment for oral defense slots: 26.06.2026 09:00 (first come first serve)

Q&A

The End – one fact sheet

- Build a data integration platform around the German Accident Atlas and Regional Atlas
- You need: Integration of official open data, Own integration script, Own database schema, Own API plus API documentation
- Main focus: correct data integration, schema design, API quality, documentation and reproducibility.
- Three out of Five Data sources: Unfallatlas, Regionalatlas Deutschland, Regionaldatenbank / GENESIS, AGS / GV-Isys, Govdata
- An Script should do reproducible imports with an extract, parse, map aggregate (ETL) into your DB
- The API should be well documented with an standard like OPENAPI
- The API should answer the mandatory questions from the category: Counts and filters, Time questions, Rates and rankings, bonus: zero cases
- A small query client is required
- An oral defense with an live demo of the questions and the tool is required
- All frameworks, languages, databases, stacks, libraries, tools are up to the student
- One task for every student, no groups are allowed
- The programm consists of an own database, a backend that processes and harmonises the data, and a web service API that exposes the stored result.
- Submission via OPAL with your ZIP file
- Mark consists 2 marks: Code and Oral defense (report depends on your study regulation)
- Submission deadline: 25.06.2026 23:59 via OPAL